

Supplementary for ch16

Abstract

This lecture note aims to provide a brief explanation of the rational expectations theory and its extension and application, known as the Ricardian equivalence theorem. Additionally, we will discuss another important theory in consumption called intertemporal consumption or the Fisher Two-Period Optimal Consumption. If you have previously studied microeconomics, you will find the content of this model to be straightforward.

1 Rational Expectation

Definition 1.1. *An approach that assumes that people optimally use all available information—including information about current and prospective policies—to forecast the future.*

According to the theory of rational expectations, a change in monetary or fiscal policy will change expectations, and an evaluation of any policy change must incorporate this effect on expectations. If people form their expectations rationally, inflation may be less inertial than it appears.

Advocates of rational expectations argue that the short-run Phillips curve does *not* accurately represent the menu of options available to policymakers. They believe that if policymakers are credibly committed to reducing inflation, rational people will understand the commitment and quickly lower their expectations of inflation. Inflation can then decline *without* a rise in unemployment and fall in output. According to the theory of rational expectations, traditional estimates of the sacrifice ratio are not useful for evaluating the impacts of alternative policies. Under a credible policy, the costs of reducing inflation may be much lower than estimates of the sacrifice ratio suggest.

In other words, with rational expectations, the Phillips curve is vertical in the short run. Since the inflation rate and unemployment rate have no trade-off relation. This means that reducing inflation doesn't have as strong of an impact on output or unemployment compared to when expectations are not considered. People anticipate future inflation and adjust their behavior

accordingly, allowing policymakers to achieve lower inflation without significant economic consequences.

The central implication of the rational expectations approach is that **people may not always get forecasts right but they do not make systematic errors.**

2 Ricardian equivalence

Definition. Consumers with **forward-looking** perspectives perceive the current government debt as future taxes. Under unchanged government expenditures, *they consider a reduction in current taxes as equivalent to an increase in future taxes.* In the case of increased government spending, they view deficit financing through taxes as equivalent to deficit financing through public debt. *Consequently, these consumers maintain consistent consumption decisions regardless of the government's tax policies and deficit financing methods.*

Example 1. The family now owns a \$100 government bond. However, the family also realizes that at the time its bond comes due, the government will find it has to raise taxes to pay off the principal and interest due on the bonds it issued to finance the deficit. So ownership of the bond does not affect the family's decisions because the value of the bond is exactly offset by the value of its implicit future tax liability. ■

There are two main theoretical objections to the Ricardo-Barro¹ proposition. *First, given that people have finite lifetimes, different people will pay off the debt than those who are receiving the tax cut today. This argument assumes that people now alive do not take into account the higher taxes their descendants will have to pay in the future.* *Second, it is argued that many people cannot borrow, and so do not consume according to their permanent income.* They would like to consume more today, but because of liquidity constraints—their inability to borrow—they are constrained to consuming less than they would want according to their permanent income. *A tax cut for these people eases their liquidity constraint and allows them to consume more.*

These theoretical disagreements mean that the Ricardo-Barro hypothesis has to be settled by examining the empirical evidence. The sharp decline of the U.S. private saving rate in the 1980s in the face of increased public deficits

¹Professor Barro, currently teaching at Harvard University, has provided a comprehensive interpretation and extension of the Ricardian equivalence theorem. As a result, the Ricardian equivalence theorem is also referred to as the "Ricardo-Barro proposition" or the "Barro-Ricardo proposition" due to Professor Barro's significant contributions.

is one piece of evidence against the proposition. Less casual empirical research continues in an attempt to settle the issue of whether the debt is wealth. We believe the evidence to date is, on balance, unfavorable to the Ricardo-Barro proposition, but we recognize that the issue has not yet been decisively settled.

3 Intertemporal Consumption Model

Let the c_i, Y_i be the consumption and income in each period, $i = 1, 2$. Suppose that utility function is increasing function as c_i increase and

$$U = U(c_1, c_2) = U(c_1) + \frac{U(c_2)}{1 + \rho}.$$

Let the time discount factor and real interest rate are ρ and r , respectively. We can write the maximization problem as follows:

$$\begin{aligned} \max_{c_1, c_2} \quad & U(c_1, c_2) = U(c_1) + \frac{U(c_2)}{1 + \rho} \\ \text{s.t.} \quad & c_1 + \frac{c_2}{1 + r} = Y_1 + \frac{Y_2}{1 + r} \end{aligned}$$

Use *Lagrange Multiplier Method* we can solve this problem as follows, let the *Lagrangian Function* is

$$\mathcal{L} = U(c_1) + \frac{U(c_2)}{1 + \rho} + \lambda(Y_1 + \frac{Y_2}{1 + r} - c_1 - \frac{c_2}{1 + r})$$

F.O.C.

$$\mathcal{L}_{c_1} = 0 \rightarrow U_1 - \lambda = 0 \rightarrow U_1 = \lambda \quad (1)$$

$$\mathcal{L}_{c_2} = 0 \rightarrow \frac{U_2}{1 + \rho} - \frac{\lambda}{1 + r} = 0 \rightarrow \frac{U_2}{1 + \rho} = \frac{\lambda}{1 + r} \quad (2)$$

$$\mathcal{L}_\lambda = 0 \rightarrow Y_1 + \frac{Y_2}{1 + r} = c_1 + \frac{c_2}{1 + r} \quad (3)$$

$$\frac{(1)}{(2)} : MRS = \frac{U_1}{U_2/(1 + \rho)} = 1 + r \Rightarrow \frac{U_1}{U_2} = \frac{1 + r}{1 + \rho} \quad (4)$$

Combine with (3) and (4) we can derive the optimal consumption c_1, c_2, s are

$$c_1 = c_1^*(r, Y_1, Y_2)$$

$$c_2 = c_2^*(r, Y_1, Y_2)$$

$$s^* = Y_1 - c_1^* = s^*(r, Y_1, Y_2)$$

From (4) we know that

$$U_1(1 + \rho) = U_2(1 + r)$$

$$r > \rho \rightarrow c_1^* < c_2^*$$

$$r < \rho \rightarrow c_1^* > c_2^*$$

$$r = \rho \rightarrow c_1^* = c_2^*$$

Therefore, we have the fact that the optimal consumption decision is **equal consumption** in each period which occur at $r = \rho$.